Ateneo de Zamboanga University
College of Science, Information Technology, and Engineering
COMPUTER SCIENCE DEPARTMENT

Research Ethics Review Application and Management

System Software Specification (SRS) Document

In partial fulfillment of the requirements in
Web Systems and Technologies

Team HASA
WEBSYSTECH - B

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Table of Contents

I. PLANNING 1

A. Project Context 1

1. About the Organization 1

2. Problems/Issues/Challenges Identified 1

A. Decentralized Information and Workflow 1

B. Unregulated Payment Collection Leading to Revenue Loss 1

C. Unreliable Applicant Communication Channels 2

D. Physical Storage Limitations and Document Degradation 2

E. Processing Bottlenecks During Peak Application Periods 2

3. Objectives 3

A. Main Objective 3

B. Specific Objectives 3

4. Significance of the System to be Developed 3

❖ Centralization and Data-Driven Decision Making (Addresses Decentralized Information): 4

❖ Financial Security and Policy Enforcement (Addresses Unregulated Payment Collection): 4

❖ Enhanced Transparency and Communication (Addresses Unreliable Communication Channels): 4

❖ Digital Archiving and Space Optimization (Addresses Physical Storage Limitations): 4

❖ Workflow Automation and Increasing Processing Speed (Addresses Processing Bottlenecks): 4

5. Planning Method 5

Phase 1: Requirements Gathering and Continuous Consultation 5

Phase 2: High-Fidelity UI/UX Prototyping (Figma) 5

Phase 3: System Development and Technology Integration 5

Phase 4: Modular Review and Iterative Alignment 6

II.ANALYSIS 7

A. Conceptual Data Modeling 7

1. ERD and Narrative 7

❖ APPLICANT and RESEARCH APPLICATION (submits) 7

❖ RESEARCH APPLICATION and PAYMENT (requires) 7

❖ PAYMENT and ADMIN (Verified by) 8

❖ RESEARCH APPLICATION and RESEARCH DOCUMENT (Contains) 8

❖ RESEARCH APPLICATION and NOTIFICATION (Generates) 8

2. Business Rules 8

B. Software Requirements 9

1. Functional Requirements 9

❖ Authentication and Role-Based Authorization 9

❖ Applicant Dashboard 9

❖ Admin Dashboard & Statistics Tracker 9

❖ Research Applications Status Updater 9

❖ Payment Verification Module (Proof of Payment) 9

❖ E-Archiving and Documentary Repository 9

❖ Searching and Filtering System 9

- ❖ Notification and E-Mail Distribution System 9
2. Non-Functional Requirements 10
- ❖ Performance 10
- ❖ Security 10
- ❖ Usability 10
- ❖ Scalability 10
- ❖ Maintainability 10
3. Constraints/Assumptions 10
- ❖ Tools and Technologies Constraint 10
- ❖ Budget Constraints 10
- ❖ Time Constraints 11
- ❖ Assumptions 11
  - ➢ Client Availability 11
  - ➢ Payment Process 11
  - ➢ Infrastructure 11
  - ➢ Email Capability 11

C. Technical Requirements 12

1. Tools 12
- ❖ Front-End Technologies & UI Libraries: 12
- ❖ Back-End Language: 12
- ❖ Database: 12
- ❖ Local Server Environment: 12

2. Hosting 12

3. Development Operations 12
- ❖ Designing & Prototyping: 12
- ❖ Code Editor: 13
- ❖ Version Control & Collaboration: 13

4. Operating Environments 13
- ❖ Operating Systems: 13
- ❖ Platforms: 13
- ❖ Web Browsers: 13

I. PLANNING

A. Project Context

1. About the Organization

The client for this web system project is the Ethics Review Board of the Ateneo de Zambonaga University (AdZU), specifically focusing on the Research Ethics Committee.

The primary mandate and purpose of the AdZU Ethics Review Board is to safeguard and uphold the ethical standards in academic and professional studies. The organization itself is responsible for meticulously reviewing research proposals and providing official ethics certifications for studies conducted both inside and outside the premises of the University.

Structurally, the board is composed of a multidisciplinary panel of experts from various fields of study who specialize in human research. This diverse composition ensures a comprehensive, unbiased, and rigorous evaluation of the ethical implications, methodologies, and safety protocols of every submitted study.

The specific process this project will be focusing on is the core administrative and evaluation workflow managed by the Research Ethics Committee with an additional Statistics Tracker of Research Applications depending on if they are Approved, Reject, or Withdraw by the Researchers themselves, and other statistical data entry such as amount of Research Applications processed Monthly and many more. This encompasses the entire lifecycle of an ethics review application, from the initial submission of research documents by internal or external applicants, through payment of fees and the manual routing of files among committee members, up to the final deliberation and issuance of the ethics certification.

2. Problems/Issues/Challenges Identified

A. Decentralized Information and Workflow

- **a. Current Process** - The Ethics Review Board currently manages data, records, and tasks across multiple separate platforms and manual logs.
- **b. Why it fails** - The information is not gathered at a singular point which leads to scattered information, this scattered information then goes through multiple cross referencing of different files and consultation of various personnel to check the status of an application. It heavily relies on manual work hand-offs between the different stages of the application.
- **c. Real Impact** - This leads to massive inefficiencies to the system, highly probable redundant date entries, and a high risk of lost or conflicting information. It wastes precious time that can be utilized to further streamline the process of the application and its status and makes management have a hard time on the logistics to get a clear picture of the real-time overview of ongoing Research Applications or Protocols.

B. Unregulated Payment Collection Leading to Revenue Loss

- **a. Current Process** - The current policy of the Ethics Review Board in terms of payment for the Research Applicants is that they can pay it at any point

of time before, during, or after the Research Application has already been processed.

b. **Why it fails** - The "pay anytime" model does not enforce a mechanism or gateway into getting the Researcher Applicants to pay due to no incentive since the Application itself will still get processed even without payment.
c. **Real Impact** - The organization suffers financial losses due to their services not getting paid and being "ghosted" by the Research Applicants even after their Application has been processed.

C. Unreliable Applicant Communication Channels

a. **Current Process -** Staff must manually reach out to applicants regarding errors, missing requirements, or status updates, relying on the initial contact information that is given by the Research Applicants.
b. **Why it fails -** Contact details are often than not outdated, misspelled, or ignored entirely by the applicants. Without a centralized portal in which the applicants themselves will be able to view the current status of their applications, communication becomes a one-sided effort.
c. **Real Impact -** Applications become bottlenecked by how fast the staff can reach out to the Researchers. Papers pile up whilst staff are waiting for the responses of the applicants inflating the overall processing timeline and causing unwarranted frustration on both ends.

D. Physical Storage Limitations and Document Degradation

a. **Current Process -** The organization requires, processes, and archives physical hard copies of the submitted Research Applications in designated storage areas inside of the office.
b. **Why it fails -** Physical storage is finite and can quickly pile up as the volume of Research Application Submissions grows. Moreover, paper documents are difficult to index, time-consuming to retrieve, and extremely vulnerable to physical damages (e.g. pests, moisture, misplacement).
c. **Real Impact -** The office environment will lead to a lot of clutter and can induce disorganization. Retrieving past research for reference will lead to slow, laborious tasks, and the organization incurs unnecessary costs for maintaining and expanding physical storage spaces.

E. Processing Bottlenecks During Peak Application Periods

a. **Current Process -** Review and processing of Research Applications rely on a heavy, multi-step manual workflow handled by a limited number of staff members.
b. **Why it fails -** Manual systems tend to be rigid and cannot scale accordingly to meet sudden urges in demand. When Research Applications volume spikes (e.g. Academic Deadlines, Submission Deadlines), the staff will be overwhelmed by the sheer amount of physical and administrative processes required on each paper.
c. **Real Impact -** This will result in massive backlogs and severe delays in each processing time. The quality of service to the Research Applicants will drop, staff will experience burnout, and Researchers are left dissatisfied with the extended waiting periods induced.

3. Objectives

A. Main Objective

- To design, develop, and implement a centralized, web-based Ethics Review Board Management System that streamlines the end-to-end processing of Research Applications, digitizes the Physical Document Storage of the Ethics Review Board, Enforcement of Financial Policies relating to the Research Applications, and Facilitate seamless communication between the administrative staff and the Research Applicants.

B. Specific Objectives

a. **To develop a centralized database and workflow dashboard** that integrates all Research Application data into a single platform, providing management with real-time tracking of ongoing protocols, tracking of statistical data of Research Applications (e.g. Accepted Research Applications, Rejected Research Applications, Withdraw Research Applications), and elimination the need for manual cross-referencing and redundant data entry.

b. **To implement a structured payment tracking module** that requires researchers to submit proof of payment (or settle fees through a gateway) as a mandatory prerequisite before their application can advance to the evaluation stages, effectively preventing revenue loss and "ghosting".

c. **To create a self-service applicant portal with automated notification** that allows researchers to view the real-time status of their applications and enables staff to easily flag errors or missing requirements without relying on outdated manual contacting methods.

d. **To establish a digital repository and e-archiving system** that allows for secure uploading, automated indexing, and rapid retrieval of soft-copy Research Application Documents, eliminating the reliance of physical storage areas and protecting files from physical degradation.

e. **To automate routine administrative task hand-offs** between different stages of the review process, enabling the organization to handle high-volume spikes during academic deadlines and submission deadlines efficiently, thereby reducing the burnout of staff and eliminating processing backlogs.

4. Significance of the System to be Developed

The proposed web-based Ethics Review Board Research Ethics Review Application and Management is of critical importance to the Ateneo de Zamboanga University (AdZU) Research Ethics Committee. By transitioning from a fragmented, de-centralized, physical, and manual framework to a centralized streamlined digital ecosystem, the system fundamentally transforms how ethics reviews are managed. It directly resolves existing operational bottlenecks and introduces a more resilient, flexible, and scalable infrastructure.

Specifically, the system addresses the identified organizational challenge in the following ways:

- **Centralization and Data-Driven Decision Making (Addresses Decentralized Information):**
  - By establishing a unified database and workflow dashboard, the system completely eliminates the need for manual cross-referencing and redundant data entry. Furthermore, the integration of a Statistics Tracker (Monitoring of approved, rejected, and withdrawn applications, alongside monthly processing volumes) provides management with a clear, real-time overview of board operations conducted in the organization. This greatly improves the accuracy and empowers the administration to make informed, data-driven decisions regarding logistics and resource allocation.
- **Financial Security and Policy Enforcement (Addresses Unregulated Payment Collection):**
  - The system directly resolves the loophole of the current "pay anytime" model of the current system. By implementing a structured payment tracking module that acts as a mandatory gateway, an application will not be able to advance to the evaluation stages without proof of payment. This effectively eliminates applicant "ghosting" and secures the organization's revenue streams.
- **Enhanced Transparency and Communication (Addresses Unreliable Communication Channels):**
  - By replacing outdated, manual outreach methods with a self-service applicant portal, the system drastically improves operational transparency. Researchers are empowered to check their real-time status autonomously, while staff can instantly flag errors, mistakes, or any missing requirements that the Research Applicants need to address to continue along the evaluation stages via automated notifications. This active communication loop eliminates waiting times and mutual frustration, fostering a much smoother applicant experience.
- **Digital Archiving and Space Optimization (Addresses Physical Storage Limitations):**
  - The implementation of a secure e-archiving repository shifts the organization away from its reliance on physical documents. This directly addresses the issue of finite physical storage areas, and eliminates the risk of physical document degradation (from pests or moisture), and turns slow, laborious manual file retrieval into an instantaneous digital search.
- **Workflow Automation and Increasing Processing Speed (Addresses Processing Bottlenecks):**
  - To mitigate severe backlogs during peak academic submission periods, the system automates routine task hand-offs between committee members. By the removal of the heavy lifting of manual file routing, the system can effortlessly scale to meet sudden urges in demand. This will drastically increase the overall processing speed, protect committee staff members from potential excessive burnout, and ensure that researchers receive high-quality, timely service.

Ultimately, the development of this web-based system guarantees a faster, highly accurate, and transparent ethics review process. It safeguards the organization's financial and physical resources while elevating the standard of service provided to the research community.

5. Planning Method

To ensure that our team will be able to successfully deliver the web-based system named Research Ethics Review Application and Management System to the Ethics Review Board, the team decided to employ **Iterative Prototyping and User-Centered Design Approach**. This methodological process emphasizes the continuous client collaboration, visual planning, and modular development to guarantee the final system precisely meets the needs of the AdZU Research Ethics Committee.

The project will be managed and executed through the following structured phases below:

Phase 1: Requirements Gathering and Continuous Consultation

The project will initiate regular interviews and meetings directly with the AdZU Ethics Review Board. These sessions are critical for the deep understanding of their specific administrative workflows, statistical tracking requirements, and distinct user journeys of both the committee members and the research applicants.

Phase 2: High-Fidelity UI/UX Prototyping (Figma)

Before any actual coding is done, the team will develop a comprehensive, interactive prototype using Figma. The prototype will visually simulate the design, layout, and user experience of the entire web-based system. It will map out exactly how the interface responds to various processes, such as the applicant portal dashboard, the payment tracking gateway, and the e-archiving search functionality. The team will present this Figma Prototype to the client for rigorous review. Only after securing official client confirmation on every page, process flow, and required module will the project advance to the development stage of the web-based system.

Phase 3: System Development and Technology Integration

Upon the approval of the prototype, the team will now transition to the technical development of the web-based system. The application will be engineered using a robust technology stack:

- **Front-End & Structure:** HTML and CSS will be utilized for the creation of the foundation and design of the Web-Based System and will integrate the usage of libraries such as **React** for Javascript and **Bootstrap** for Front-End Development Framework. These Libraries and Frameworks are chosen to streamline the development process, keep the codebase clean and modular, and eliminate time-consuming “boilerplates” or burner code, ensuring a highly responsive and user-friendly interface.
- **Back-End & Database:** The system’s server-side logic and automated hand-offs will be powered by **PHP**. Data management, including the centralized database for research protocols and statistical tracking, will be handled using **MariaDB** hosted via a local **XAMPP** environment during the development and testing stages.

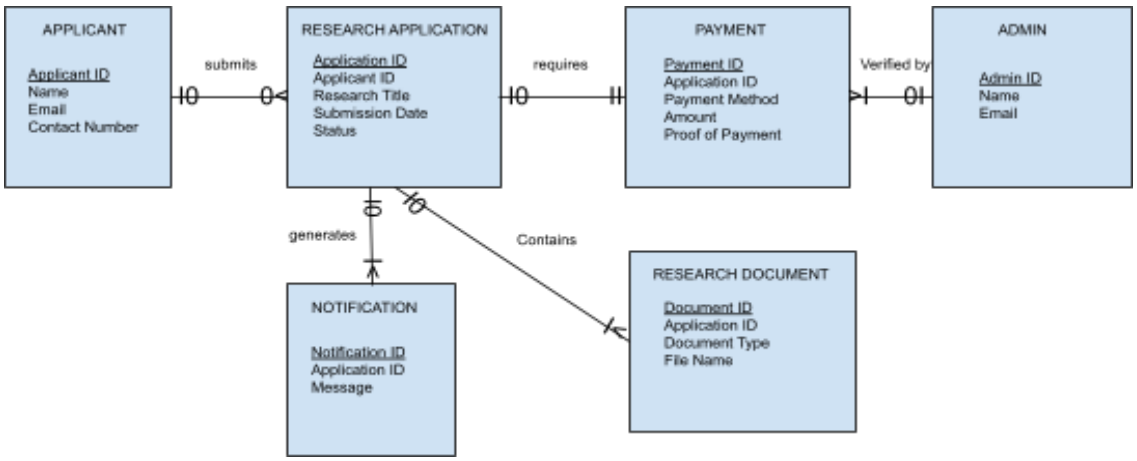
Phase 4: Modular Review and Iterative Alignment

To manage the project efficiently and prevent final-stage misalignments with the client which is the Ethics Review Board of Ateneo de Zamboanga University, development will be broken down into distinct modules (e.g. Payment Module, Automated Notifications System, Admin Dashboard, Applicant Dashboard). Upon the completion of each individual module or a cluster of modules (if the modules themselves are relatively small or take up a minor part of an encompassing process), the team will hold a demonstration meeting with the client. This continuous feedback loop allows the organizations to dry-run the features in real-time, providing additional feedback, and allow the developers to implement improvements iteratively before moving on to the next segment of the system.

II.ANALYSIS

A. Conceptual Data Modeling

1. ERD and Narrative



TThe conceptual data model shows the main entities needed for the **Research Ethics Review Application and Management System**. The six main entities are **APPLICANT, RESEARCH APPLICATION, PAYMENT, ADMIN, RESEARCH DOCUMENT,** and **NOTIFICATION**. These entities are connected based on how the application process works.

- **APPLICANT and RESEARCH APPLICATION (submits)**

  An Applicant is a researcher who submits a research application. One applicant can submit zero, one, or many research applications. Each Research Application is submitted by exactly one applicant.

- **RESEARCH APPLICATION and PAYMENT (requires)**

  A Research Application may have zero or one payment because an application may still be unpaid or in draft status. Each payment is connected to exactly one research application.

- **PAYMENT and ADMIN (Verified by)**

  An admin is responsible for checking and verifying payments. One admin can verify zero, one, or many payments. A payment can be verified by zero or one admin, since it may still be waiting for verification.

- **RESEARCH APPLICATION and RESEARCH DOCUMENT (Contains)**

  A Research Application can contain zero, one, or many research documents. Each Research Document belongs to exactly one research application.

- **RESEARCH APPLICATION and NOTIFICATION (Generates)**

  A Research Application can generate zero, one, or many notification**s** for things such as status updates, missing requirements, or other application information. Each Notification is connected to exactly one research application.

2. Business Rules

Relationship: **Submits (APPLICANT – RESEARCH APPLICATION)**

- An Applicant can submit zero, one, or many Research Applications.
- Each Research Application is submitted by exactly one Applicant.

Relationship: **Requires (RESEARCH APPLICATION – PAYMENT)**

- A Research Application can require zero or one Payment.
- Each Payment is connected to exactly one Research Application

Relationship: **Verified by (PAYMENT – ADMIN)**

- A Payment can be verified by zero or one Admin.
- An Admin can verify zero, one, or many Payments.

Relationship: **Contains (RESEARCH APPLICATION – RESEARCH DOCUMENT)**

- A Research Application can contain zero, one, or many Research Documents
- Each Research Document belongs to exactly one Research Application

Relationship: **Generates (RESEARCH APPLICATION – NOTIFICATION)**

- A Research Application generates zero, one, or many Notifications.
- Each Notification is generated by exactly one Research Application

B. Software Requirements

1. Functional Requirements

- **Authentication and Role-Based Authorization**
  - The system will provide secure login portals tailored to user roles. It will differentiate access levels between Research Applicants and Ethics Review Board (ERB) Administrations, ensuring that Researchers can only view their own submissions while admins have access to the overarching management tools and all submitted documents.
- **Applicant Dashboard**
  - A self-service portal where researchers can submit new applications, upload required soft-copy documents, and view real-time status updates of their ongoing ethics review. This module will eliminate the need for manual inquiries by providing applicants with immediate visibility into their application's progress.
- **Admin Dashboard & Statistics Tracker**
  - A centralized command center for ERB staff and admins to manage incoming and ongoing applications. This dashboard will include a built-in statistics tracker that visually displays key data metrics, such as the volume of applications processed monthly, and the total counts of Approved, Rejected, and Withdrawn Applications.
- **Research Applications Status Updater**
  - A workflow module that allows administrators to manually or automatically advance an application through its specific evaluation stages. As the application moves from initial submission to final deliberation, this updater tracks the current stage, automates administrative hand-offs, and triggers necessary alerts.
- **Payment Verification Module (Proof of Payment)**
  - A mandatory gateway that enforces the organization's financial policy. Before an application can advance to the evaluation stage, applicants must upload their official proof of payment (secured via the AdZU Finance Department). Admins will use this module to verify and approve the payment, securely logging the transaction and preventing revenue loss.
- **E-Archiving and Documentary Repository**
  - A secure digital storage system where all uploaded research papers, protocols, and forms are indexed and saved. This module completely replaces the physical filing system, protecting documents from physical degradation and allowing for instant retrieval.
- **Searching and Filtering System**
  - A robust query tool allowing administrative staff to instantly retrieve specific past or ongoing research applications. Users will be able to filter records based on parameters such as date, application status, researcher name, or document category.
- **Notification and E-Mail Distribution System**
  - An automated alert system directly into the dashboard and connected via email. It will instantly distribute critical communications such as Progress, Reports, Major/Minor Amendments, Letters of Decision, Final Report

Letters, and official Certifications/Clearances to both the applicants' and the ERB Administrators' work emails.

2. Non-Functional Requirements

- **Performance**
  - The system must handle document uploads and status queries efficiently, with page load times expected to be under 3-5 seconds on standard broadband connections. The dashboard and statistics tracker must render data visualizations without causing noticeable lag/artificing to the user experience
- **Security**
  - User data and intellectual property (research papers) must be protected using standard web security protocols. This includes password hashing for user accounts, secure session management, and the use of parameterized queries (via PHP/MariaDB) to prevent SQL Injection attacks. Document access will be strictly restricted based on user roles.
- **Usability**
  - The user interface must be highly intuitive and responsive without having a manual or lack thereof a manual, utilizing React and Bootstrap to ensure the system is accessible across various devices (desktops, laptops, and tablets). The layout will strictly adhere to the pre-approved Figma Prototype to guarantee that users require minimal or no training to navigate the application.
- **Scalability**
  - While initially designed for current operational volumes, the underlying database architecture (MariaDB) must be structured to support a continuously growing repository of research applications, user accounts, and digital files without performance degradation over the coming years.
- **Maintainability**
  - The source code will follow clean, modular programming principles. By utilizing component-based development in React and structured server-side logic in PHP, the codebase will be easy to read, debug, and upgradeable by future development teams or IT personnel.

3. Constraints/Assumptions

- **Tools and Technologies Constraint**
  - Development is strictly confined to a specific technology stack: HTML, CSS, React, and Bootstrap for the front-end interface, with PHP and MariaDB handling the backend and database management. The system will be developed and initially tested using a local XAMPP Environment. Design and Prototyping are limited to Figma.
- **Budget Constraints**
  - The project operates on a zero-cost budget. All development, design, and testing rely exclusively on free, open-source software, libraries, and frameworks. Premium third-party APIs, paid plugins, or enterprise-tier cloud hosting services cannot be utilized during the development phase.

- **Time Constraints**
  - The project is bound by a strict, non-negotiable development timeline. All phases from initial client consultation and Figma Prototyping to final coding, module testing, and system delivery must be fully executed within the span of the whole 1st Semester of the Academic Year 2026 - 2027.
- **Assumptions**
  - **Client Availability**
    - it is assumed that the client (ERB members) will be available for regular modular review meetings to provide timely feedback and approvals.
  - **Payment Process**
    - It is assumed that the AdZU Financial Department will continue to issue standard, verifiable proof-of-payment receipts that applicants can photograph or scan to upload into the system.
  - **Infrastructure**
    - It is assumed that both ERB Staff and Research Applicants possess reliable internet connectivity and modern web browsers capable of rendering React-based Applications.
  - **Email Capability**
    - It is assumed that a standard, free SMTP server or basic PHP mailer configuration will be sufficient for handling the automated email distribution system without encountering severe spam-filtering issues.

C. Technical Requirements

1. Tools

The development of the web-based Research Ethics Review Application and Management System will utilize the following Technology Stack, chosen for its reliability, modularity, and rapid development capabilities:

- **Front-End Technologies & UI Libraries:**
  - The user interface will be utilizing **HTML**, **CSS**, and **React**. React is selected due to its component-based architecture, which is ideal for creating dynamic, interactive interfaces like the Applicant and Admin Dashboards. **Bootstrap** on the other hand is chosen to be primarily utilized as the primary CSS Framework to ensure rapid, clean, and mobile-responsive styling without the need to write excessive custom code for the design elements.
- **Back-End Language:**
  - **PHP** will be used to handle the server-side logic. It will encompass the processes from form submissions, manage user authentication, execute the automated hand-offs in the review workflow, and handle the automated email notification system.
- **Database:**
  - **MariaDB** will serve as the relational database management system. It is highly capable of securely storing, indexing, and linking complex data structures, such as user credentials, research application statuses, statistical logs, and digital file directories.
- **Local Server Environment:**
  - **XAMPP** will be utilized to provide a seamless local development environment, bundling the necessary Apache Server, MariaDB, and PHP components together

2. Hosting

During the initial coding, testing, and modular review phases, the system will be hosted **locally** on the developers' machines utilizing the XAMPP Environment. Once the system is fully developed and prepared for final deployment and live demonstration to the client, it will be hosted and routed through the **Free tier of Cloudflare**. This ensures that the final web application is securely accessible via the internet, benefits from baseline performance optimizations, and provides a stable live environment for the users without incurring external infrastructure costs.

3. Development Operations

To maintain an organized and collaborative workflow, the team will be utilizing the following operational tools:

- **Designing & Prototyping:**
  - **Figma** will be used exclusively for the initial high-fidelity prototyping phase, allowing the team to map out layouts, module interactions, and overall UI/UX before any code is written.

- **Code Editor:**
  - **Visual Studio Code (VS Code)** will serve as the primary Integrated Development Environment (IDE) for the entire development team, chosen for the robust extensions, terminal integration, and excellent support for both **React** and **PHP**.
- **Version Control & Collaboration:**
  - The team will be managing the system's codebase using **Git** and **GitHub** as the central repository. To accommodate different development workflows, team members will interact with the repository using a mix of tools: some will be utilizing **Git Bash** via the command-line interface for granular control over commits and branches, while others will be utilizing **GitHub Desktop** for a streamlined, visual approach to version control. Node Package Manager (npm) will be utilized to handle necessary React dependencies.

4. Operating Environments

- **Operating Systems:**
  - The web application is primarily optimized for the Windows operating system, aligning with the standard office hardware and administrative infrastructure currently utilized by the AdZU Ethics Review Board staff.
- **Platforms:**
  - The system is designed to be fully responsive and accessible across both **Desktop/Laptop** and **Mobile** platforms. This flexibility ensures that administrators have the screen space necessary for complex data management on desktops, while applicants have the convenience of checking their research application status via their mobile devices.
- **Web Browsers:**
  - The systems will be compatible with all modern, updated web browsers (e.g. Google Chrome, Mozilla Firefox, Microsoft Edge, Safari) that supports **HTML5**, **CSS3**, and **React-based** dynamic rendering. No specialized software installations are required for end-users.